

# Spacecraft Pursuit-Evasion with Fictitious Play over Mixed Trajectory Strategies

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**Abstract**—Spacecraft pursuit-evasion games are traditionally formulated as deterministic differential games; however, pure-strategy solutions are often exploitable or non-existent in constrained orbital environments. This paper proposes a framework for non-cooperative orbital games by projecting continuous control problems into a discrete set of candidate trajectories. By employing mixed strategies and Fictitious Play (FP), we enable an agent to adapt to an opponent’s strategy in real-time. We evaluate our approach through Monte Carlo simulations in a three-dimensional orbital environment, showing that FP strategies outperform locally optimal but naive strategies by a factor of 2.7 in game value and 3.22 in inter-player distance. This framework demonstrates the significant strategic edge that learning provides for mission success.

## I. INTRODUCTION

Multi-agent systems are essential in modern space operations. Distributed communication networks, GPS and imaging constellations, and rendezvous and docking procedures all rely on agents that must predict and coordinate with others to achieve mission success. Historically, these interactions have been treated as cooperative multi-agent problems, where all actors share a common objective. Such scenarios are typically solved as one-sided optimal control problems that determine the motion of all agents simultaneously. However, the rise of international competition has introduced an era of adversarial actors in the space domain. In this non-cooperative landscape, an agent must fulfill its own mission requirements, such as maintaining a specific orbital slot, while proactively mitigating risks posed by other agents with competing objectives.

Game theory provides a rigorous mathematical framework for navigating these strategic interactions. This approach treats pursuit-evasion as a two-sided optimization problem, seeking a Nash equilibrium: a state where neither agent can reduce their respective costs by unilaterally changing their control strategy. While these classical methods provide a theoretical foundation for optimal play, they often struggle with the nonlinear complexities and high dimensionality of orbital dynamics [1] [2]. Another limitation is their reliance on deterministic, or “pure,” strategies. In a competitive environment, a deterministic strategy is inherently exploitable; if an opponent can predict a specific trajectory, they can utilize this information to counteract it [3]. Furthermore, existing literature frequently assumes the pursuer possesses superior fuel and agility, effectively guaranteeing capture [4] [5], though in practice this may not be true.

Motivated from recent advances in robotics, we explore the use of mixed strategies, where players optimize over a finite set of candidate trajectories [3] [6], alleviating issues arising from deterministic play. Additionally, by employing a framework for profiling agent behavior called fictitious play (FP), agents can adapt to an opponent’s real-time actions [7]. In this work, we optimize over a discrete set of candidate trajectories to determine best-response strategies. Our primary contributions are as follows:

- 1) We develop a methodology to project the continuous control pursuit-evasion game into a discrete set of candidate trajectories around a reference orbit.
- 2) We utilize fictitious play, allowing an agent to profile an opponent’s strategy in real-time.
- 3) We provide a Monte Carlo table of stationary results by pitting various strategy types against one another: Greedy, Mixed, Random, and Fictitious Play.

## II. RELATED WORK

Our work builds upon contributions in numerical methods for spacecraft trajectory optimization, game-theoretic planning, and multi-agent behavior reasoning.

### A. Pursuit-Evasion as Optimal Control

The foundation of spacecraft pursuit-evasion (PE) lies in differential game theory, typically formulated via Hamilton-Jacobi-Isaacs (HJI) to identify saddle-point equilibria [8]. Orbital PE research often relies on linearized Hill-Clohessey-Wiltshire (HCW) dynamics for proximity operations [9] [10] [11]. Semi-direct numerical methods have been combined with analytical necessary conditions identify equilibria in 3D scenarios [5]. More recently, high-fidelity strategies have been developed by bifurcating the pursuit process into distinct navigation and acceleration phases, allowing for rigorous analytical thresholds for capture within the HJBI framework [12].

Despite these advances, the “curse of dimensionality” remains a primary bottleneck, as gridding a 12-dimensional state space for two-body games is computationally prohibitive for real-time applications. Efforts have focused on reducing this burden through analytical strategies using Tschauner-Hempel equations [1] and reachability analysis [2] [13]. Linear-quadratic approximate methods have enabled the study of adversarial maneuvers in complex regimes, including zero-sum games in cislunar space [14] and collision avoidance [15].

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### B. Mixed Strategies and Type-Based Reasoning

A significant limitation in classical PE literature is the reliance on deterministic (pure) strategies. In highly constrained orbital environments, where  $\Delta V$  budgets are comparable, a pure-strategy Nash equilibrium may not exist, or the predicted trajectory may be exploited by an opponent [6]. Recent advances in robotics have addressed this by “lifting” continuous games into a discrete space of candidate trajectories or motion primitives [3], which can lead to matrix-game formulations that reduce the online computational burden of trajectory optimization. Leveraging learning dynamics such as fictitious play can ensure strategy convergence [16], [17], and such mixed-strategy approaches have been successfully applied to multi-spacecraft interception [6] and games involving safety-constrained robot interactions [18].

By maintaining a probability distribution over a set of possible choices, an agent can profile an opponent’s strategy in real-time. In this work, we project the pursuit-evasion game into a discrete trajectory space defined by the orbital bandwidth and apply this update mechanism to transition from a generic game-theoretic solution to a tailored best-response [19], [20]. We provide a framework for adaptation in non-stationary gameplay, specifically within the dynamics of orbital pursuit-evasion.

### III. PROBLEM FORMULATION

Consider a system of a pursuer  $P$  and an evader  $E$  operating in an orbital dynamics environment. The state of each agent  $i \in \{P, E\}$  is defined as  $\mathbf{x}_i(t) = [\mathbf{r}_i^\top(t), \mathbf{v}_i^\top(t)]^\top \in \mathbb{R}^6$ , where  $\mathbf{r}_i$  and  $\mathbf{v}_i$  represent Cartesian position and velocity in the Earth-Centered Inertial (ECI) frame at time  $t \in \mathbb{R}$ . The discrete-time evolution of the state is governed by

$$\mathbf{x}_i(t_{k+1}) = F(\mathbf{x}_i(t_k), \mathbf{u}_i(t_k)), \quad (1)$$

where  $F : \mathbb{R}^6 \times \mathbb{R}^3 \rightarrow \mathbb{R}^6$  transitions the state from  $t_k$  to  $t_{k+1}$ . The control input  $\mathbf{u}_i(t_k) = \Delta \mathbf{v}_i \in \mathbb{R}^3$  is modeled as an impulsive maneuver, or  $\Delta V$  velocity change applied at  $t_k$ . This discrete-step impulsive model follows the **Sims-Flanagan method of low-thrust trajectory optimization for orbital dynamics** [21].

#### A. Objective Functions

The agents have strictly competing objectives, forming a zero-sum differential game. The evader seeks to remain within an **orbital bandwidth  $\mathcal{B}$  centered on a nominal reference trajectory**, while maximizing its distance from the pursuer. Conversely, the pursuer attempts to minimize this distance to achieve interception. We define the **stage cost  $\mathcal{C}(\cdot)$**  at time  $t_k$  as the relative distance between agents

$$\mathcal{C} = \|\mathbf{r}_P(t_k) - \mathbf{r}_E(t_k)\|_2. \quad (2)$$

The total objective function over an  $L$ -step finite horizon is

$$J(\mathbf{U}_P, \mathbf{U}_E) = \sum_{k=0}^{L-1} \mathcal{C}(t_k, \mathbf{r}_E, \mathbf{r}_P, \mathbf{u}_E, \mathbf{u}_P) \quad (3)$$

where the control sequence  $\mathbf{U}_i = \{\mathbf{u}_i(t_0), \dots, \mathbf{u}_i(t_{L-1})\}$  governs the evolution of each player’s positions,  $\mathbf{r}_E$  and  $\mathbf{r}_P$ .

### B. Model-Predictive Control Games

Model-Predictive Control (MPC) can be used to solve pursuit-evasion games over a shifting horizon. At each epoch, the agents solve a **minimax optimization problem**

$$\begin{aligned} & \min_{\mathbf{U}_P} \max_{\mathbf{U}_E} J(\mathbf{U}_P, \mathbf{U}_E) \\ & \text{subject to } \mathbf{x}_i(t_{k+1}) = F(\mathbf{x}_i(t_k), \mathbf{u}_i(t_k)) \\ & \mathbf{x}_i(t_k) \in \mathcal{F} \\ & \|\mathbf{u}_i(t_k)\|_2 \leq \Delta V_{\max, i} \end{aligned} \quad (4)$$

where  $\mathcal{F}$  represents the set of feasible states within the orbital bandwidth  $\mathcal{B}$ . A saddle point solution to this problem is a **Nash Equilibrium  $(\mathbf{U}_P^*, \mathbf{U}_E^*)$** , satisfying

$$J(\mathbf{U}_P^*, \mathbf{U}_E) \leq J(\mathbf{U}_P^*, \mathbf{U}_E^*) \leq J(\mathbf{U}_P, \mathbf{U}_E^*). \quad (5)$$

Each agent then begins to execute its equilibrium strategy and moves for a short time period before repeating the finite-time trajectory game via (4).

However, in highly constrained orbital environments, a pure-strategy Nash equilibrium in the continuous control space may not exist, or may be computationally prohibitive to find in real-time [2]. Furthermore, deterministic strategies derived from the HJBI equations are susceptible to exploitation if one agent deviates from “rational” play [18].

### IV. APPROACH

To address the issues from equilibrium non-existence and deterministic strategies, we move from the continuous control space  $\mathbf{U}_i$  to a lifted game played over a discrete set of candidate trajectories. We compute the mixing weights for each strategy and implement fictitious play to enable an agent to adapt to an opponent’s strategy in real-time.

#### A. Orbital Discretization and Hexagonal Lifting

To resolve issues of non-uniqueness and to allow for the use of mixed strategies, we first discretize the reference trajectory into a total of  $K$  segments. At each segment, we can define a local reference frame to approximate the orbital bandwidth. We define basis  $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3\}$  at each node:

- $\hat{\mathbf{e}}_1$ : Aligned with the orbital velocity vector (prograde).
- $\hat{\mathbf{e}}_2$ : Points toward the center of the Earth (nadir).
- $\hat{\mathbf{e}}_3$ : The cross product of  $\hat{\mathbf{e}}_1$  and  $\hat{\mathbf{e}}_2$  (cross-track).

A six-sided polygon (hexagon) is centered on the reference trajectory within the  $\hat{\mathbf{e}}_2 - \hat{\mathbf{e}}_3$  plane. This hexagonal cross-section approximates the terminal reachable set of the evader for each horizon.

#### B. Candidate Trajectory Generation

For each planning horizon, both the pursuer and evader generate a set of six candidate trajectories, each targeting one of the hexagon’s vertices  $v_j$  for  $j = \{1, 2, \dots, 6\}$  at  $L$  segments into the future. The reachable set technically contains infinite points, but we constrain the target points to 6 for computational tractability. **For each agent’s trajectory, we set up a model-predictive control (MPC) problem with  $L$  impulsive maneuvers to reach the vertex while minimizing fuel. A Lambert solver generates a geometric solution for the**

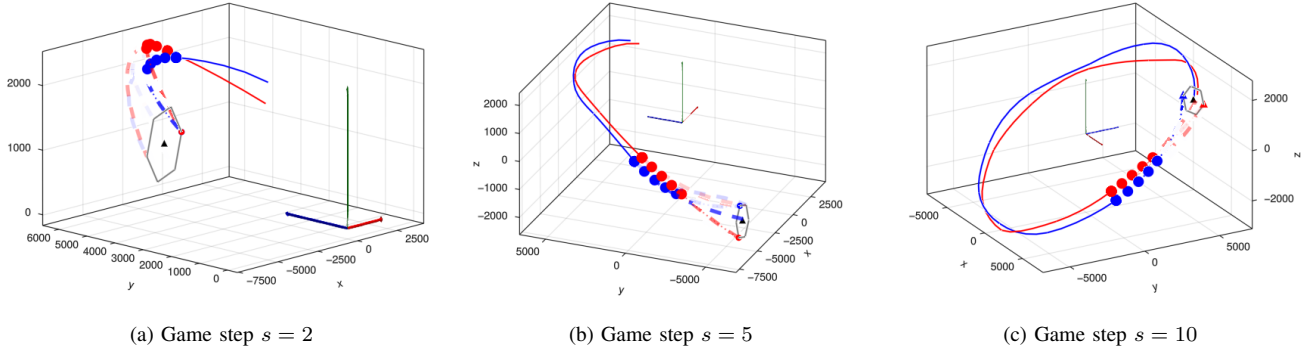


Fig. 1: **Receding-horizon model predictive game, where blue shows the evader’s path, and red represents the pursuer’s.** The inertial frame’s  $x$ ,  $y$ , and  $z$  axes are represented by the red, blue, and green arrows respectively. At each game step  $s$ , both players generate candidate trajectories for matrix game formulation and compute mixing weights to find the Nash equilibrium. The solid lines represent the traveled trajectories in past game steps. For the current step, the thickness of the dashed lines represent each candidate trajectory’s weights, and the spheres represent the chosen propagated states.

initial guess [22], and solving the MPC problem recovers the optimal control inputs for each agent  $\mathbf{u}_i$

$$\begin{aligned} \min_{\mathbf{u}_i} \quad & \sum_{k=0}^{L-1} \|\mathbf{u}_i(t_k)\|_2 + \|\Delta \mathbf{r}_i(t_L)\|_2 \\ \text{subject to} \quad & \mathbf{x}_i(t_{k+1}) = F(\mathbf{x}_i(t_k), \mathbf{u}_i(t_k)) \\ & \mathbf{x}_i(t_k) \in \mathcal{F} \\ & \|\mathbf{u}_i(t_k)\|_2 \leq \Delta V_{\max} \end{aligned} \quad (6)$$

The control input  $\mathbf{u}_i(t_k)$  represents the  $\Delta V$  impulsive maneuver at time  $t_k$ , which is constrained by  $\Delta V_{\max}$ . The final miss distance at each vertex  $v_j$  is

$$\Delta \mathbf{r}_i(t_L) = \mathbf{x}_i(t_L) - \mathbf{r}_{v_j}. \quad (7)$$

where  $\mathbf{r}_{v_j}$  is the vertex position in the local orbital frame. Now, we compute the game cost matrix for the planning horizon. For every pair of candidate trajectories  $(j, l)$ , where  $v_j$  is the evader’s target vertex and  $v_l$  is the pursuer’s target vertex, we compute the averaged stage cost

$$A_{jl} = \frac{1}{L} \sum_{k=0}^{L-1} C(t_k, \mathbf{r}_E, \mathbf{r}_P, \mathbf{u}_E, \mathbf{u}_P), \quad (8)$$

where  $C(\cdot)$  represents the stage cost

$$C = \sqrt{\|\mathbf{r}_P(t_k) - \mathbf{r}_E(t_k)\|_2 + \lambda_1 + \lambda_2(\|\mathbf{u}_P(t_k)\|_2 - \|\mathbf{u}_E(t_k)\|_2)}. \quad (9)$$

For six candidate trajectories, this results in a  $6 \times 6$  payoff matrix  $\mathbf{A}$ . Here,  $\lambda_1$  is a smoothing parameter and  $\lambda_2$  weights the relative control effort.

### C. Mixed Strategies via Linear Programming

We seek a mixed strategy to avoid exploitation of deterministic strategies by other agents. We first transform  $\mathbf{A}$  to be entry-wise positive to satisfy the positivity constraints for linear programming (LP) [8]. The optimal mixing weights are found by solving the following LP

$$\begin{aligned} \max_{\mathbf{z} \in \mathbb{R}^6} \quad & \mathbf{1}^\top \mathbf{z} \\ \text{subject to} \quad & \mathbf{A}^\top \mathbf{z} \leq \mathbf{1} \\ & \mathbf{z} \geq \mathbf{0}. \end{aligned} \quad (10)$$

We then normalize the solution to obtain the Nash optimal probability distribution  $\mathbf{w}^* \in \mathbb{R}^6$  (the “mixing weights”)

$$\mathbf{w}^* = \frac{\mathbf{z}^*}{\mathbf{1}^\top \mathbf{z}^*}. \quad (11)$$

Here, each agent can then sample from their respective  $\mathbf{w}^*$  to select a trajectory for the current horizon.

### D. Receding-Horizon Fictitious Play

The structure governing game-theoretic belief updates and model-predictive game-play nests across time scales:

- 1) Orbital Segment  $k \in \{1, \dots, K\}$ : the fundamental temporal discretization of the orbit.
- 2) MPC Horizon  $L$ : the window comprising  $L$  segments used for trajectory optimization at each step.
- 3) Game Step  $s \in \{1, \dots, S\}$ : the decision interval, spanning  $N$  orbital segments ( $N < L$ ), in which agents profile opponent behavior and execute strategies.

Let  $t_k$  denote the time at the current orbital segment  $k$  and  $s$  the current game step. Each agent generates a set of six candidate trajectories and computes a payoff matrix  $\mathbf{A}$  for the horizon  $[t_k, t_{k+L}]$ . Agents update their beliefs regarding the opponent’s strategy based on observation histories and execute the best-response mixed strategy for the game step interval  $[t_k, t_{k+N}]$ . At step  $s + 1$ , the time shifts to  $t_{k+N}$ , and the process repeats in a receding-horizon fashion.

1) *Empirical Belief Update*: At each game step  $s$ , the agent observes the opponent’s most recent vertex choice in the past. We represent this choice as  $\mathbf{g}_{s-1} \in \{0, 1\}^6$ , where the  $j$ -th entry is 1 if the opponent targeted vertex  $v_j$ , otherwise 0. The observation vector  $\mathbf{f} \in \mathbb{R}^6$  is initialized as ones, and the agent updates its cumulative observation history  $\mathbf{n} \in \mathbb{R}^6$  via

$$\mathbf{f}_s = \mathbf{f}_{s-1} + \mathbf{g}_{s-1}. \quad (12)$$

The empirical frequency estimate  $\hat{\mathbf{q}} \in \mathbb{R}^6$  is then marginalized over all past game steps to form the current belief

$$\hat{q}_i = \frac{f_j}{\sum_{m=1}^6 f_m}. \quad (13)$$

2) *Fictitious Play Strategy Exploitation*: Each player then computes the expected cost of each of its own vertices under the empirical belief  $\hat{\mathbf{q}}$  about the opponent's vertex distribution. For the evader,

$$\mathbf{c}_{\text{exp}} = \mathbf{A} \hat{\mathbf{q}}, \quad (14)$$

where the  $j$ -th entry gives the expected stage cost if the evader selects vertex  $v_j$ . For the pursuer,

$$\mathbf{c}_{\text{exp}} = -\mathbf{A}^\top \hat{\mathbf{q}}, \quad (15)$$

where the transpose reflects the reversal of row and column roles and the negation captures the zero-sum structure. Each player selects  $v^* = \arg \max \mathbf{c}_{\text{exp}}$  and executes the first  $N$  segments (one game-step duration) before generating candidate trajectories once again for the next MPC horizon.

## V. EXPERIMENTS & RESULTS

We have presented a game-theoretic framework for orbital pursuit-evasion in which fictitious play enables each player to learn and exploit the opponent's strategy in real time. In this section, we evaluate the performance of the proposed approach on a two-spacecraft pursuit-evasion scenario in low Earth orbit, quantify the advantage of fictitious play as the game horizon lengthens, and demonstrate that the belief update reliably exploits the opponent's strategy within a few orbital periods.

### A. Simulation Set-Up

We consider a pursuit-evasion scenario in low Earth orbit. The reference trajectory is a circular orbit at an altitude of 620 km (semi-major axis  $a = 6998$  km, eccentricity  $e = 0.01$ , inclination  $i = 20^\circ$ ), yielding an orbital period of approximately 97 minutes. The evader (Player 1) is initialized on this reference orbit, while the pursuer (Player 2) begins on a slightly higher orbit with semi-major axis  $a_P = 1.005 a$ , introducing a natural along-track drift between the two vehicles. Both spacecraft are modeled as point masses executing impulsive  $\Delta V$  maneuvers; no continuous-thrust or attitude dynamics are included, so the control authority of each player is characterized entirely by a per-segment velocity-change constraint of  $\Delta V_{\text{max}} = 0.1$  km/s.

At each game step, both players simultaneously select one of six target vertices arranged as a regular hexagon of radius  $R = 6.378$  km in the local orbital frame. We compute the  $\Delta V$  impulses to generate the candidate trajectories from the current state to each vertex via (6). The game is discretized into  $K = 50$  segments per orbit, and each player optimizes over an  $L = 10$ -segment horizon. The resulting  $6 \times 6$  cost matrix is computed via (8) and (9) with regularization parameters  $\lambda_1$  and  $\lambda_2 = 0.1$ . The game is zero-sum:  $\mathbf{A}_1 = -\mathbf{A}_2$ , and the mixed Nash equilibrium is computed by LP via (10) each step, with a minimum probability floor of  $10^{-4}$  on each vertex. The LP is modeled in JuMP and solved with OSQP [23], an operator-splitting solver for quadratic programs. Each game step executes for  $N = 5$  segments, and then the MPC horizon shifts to generate the next set of candidate trajectories.

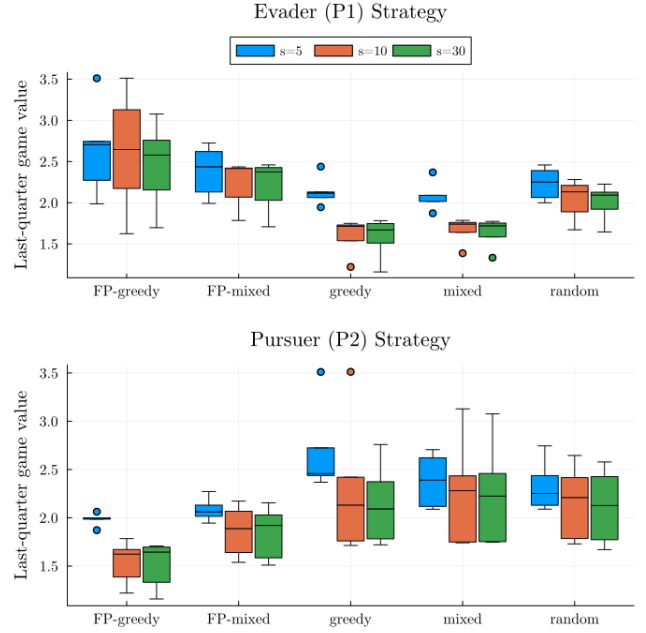


Fig. 2: **The FP-greedy strategy dominates for both players.** The box plots report game values over the last-quarter window at game steps 5, 10, and 20. Higher values are better for the evader; lower values are better for the pursuer.

Five strategies are evaluated for each player:

- 1) **Greedy**: deterministically select the vertex with the highest Nash equilibrium weight.
- 2) **Mixed**: sample a vertex from the Nash equilibrium mixed strategy at each step.
- 3) **Random**: select a vertex uniformly at random, independent of the cost matrix.
- 4) **FP-greedy**: form the empirical belief  $\hat{\mathbf{q}}$  via (13) based on opponent's past vertex choices (fictitious play) and select the best response  $\mathbf{c}_{\text{exp}}$  via (14) for each player.
- 5) **FP-mixed**: the belief update is identical to FP-greedy, but instead of always choosing the best response, sample from  $\mathbf{c}_{\text{exp}}$ .

We evaluate all 25 pairwise matchups of the five strategies across three periods:  $S = 10, 20$ , and 30 game steps, which corresponds to 1, 2, and 3 orbital periods. Each matchup is run for 50 Monte Carlo trials with identical initial orbits but independent random seeds for strategy sampling.

### B. Results

We report three classes of metrics: (i) the game value (mean stage cost), (ii) the mean inter-player distance, (iii) the cumulative  $\Delta V$  for each player. To isolate steady-state behavior from transient effects, we report last-quarter data gathered over the final 25% of the segment timeline for the game value and inter-player distance metrics.

1) *Strategy Dominance*: Figure 2 presents the mean game value averaged across all opponents, for each evader (P1) and pursuer (P2) strategy at each horizon. Higher values favor the evader, and lower values favor the pursuer.

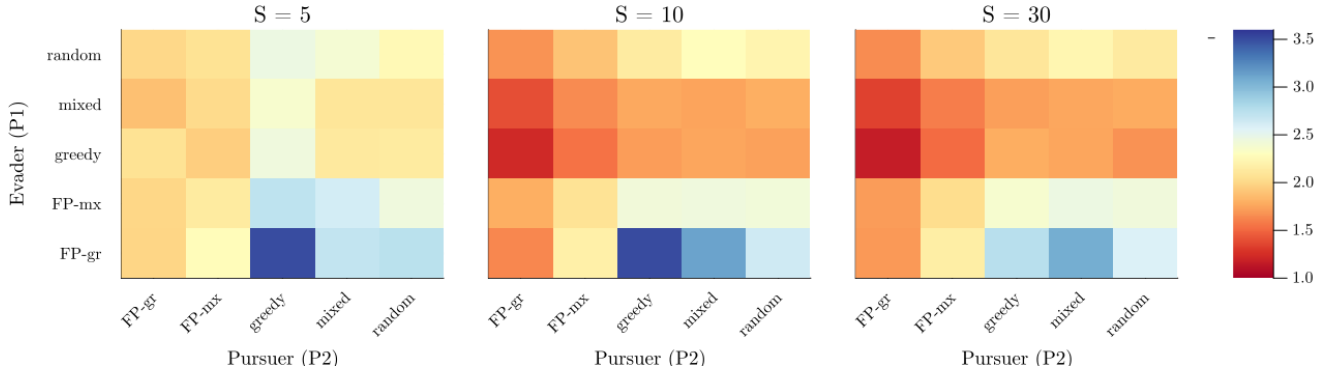


Fig. 3: **Game value heatmaps for the evader-pursuer matrix game at 5, 10, and 30 game steps.** Blue corresponds to higher game values and is better for the evader, while red is lower and better for the pursuer. The strategy pair game values are more or less uniform at 5 game steps, but the effects sharpen at 10 game steps. FP-greedy is the clear dominant strategy for the pursuer, but behavior profiling benefits both players.

TABLE I: **Game value matrix at  $s = 30$ .** Rows correspond to the evader (P1) strategy; columns to the pursuer (P2). Higher values favor the evader.

| P1 \ P2   | FP-greedy   | FP-mixed | greedy | mixed       | random |
|-----------|-------------|----------|--------|-------------|--------|
| FP-greedy | 1.70        | 2.15     | 2.76   | <b>3.08</b> | 2.58   |
| FP-mixed  | 1.71        | 2.03     | 2.37   | 2.46        | 2.43   |
| greedy    | <b>1.16</b> | 1.51     | 1.78   | 1.75        | 1.67   |
| mixed     | 1.33        | 1.59     | 1.72   | 1.75        | 1.77   |
| random    | 1.64        | 1.92     | 2.09   | 2.22        | 2.13   |

FP-greedy is the dominant strategy on both sides of the game: the best evader and the best pursuer for every strategy pair tested. The ranking is consistent: fictitious-play strategies outperform naive ones. Notably, greedy and mixed strategies perform worst for the evader, while random strategies approach the performance of FP-greedy and FP-mixed strategies. Meanwhile, for the pursuer, FP-greedy unambiguously outperforms all other strategies, exemplifying the benefit of unpredictability for the evader.

2) *Strategy-Pair Matchups*: The  $5 \times 5$  game-value matrix (Table I) and distance matrix (Table II) further reveal the strategic landscape for each player. The performance gap is significant: the evader’s most successful pairing (FP-greedy vs. mixed pursuer, 3.08) exceeds its least successful (greedy vs. FP-greedy pursuer, 1.16) by a factor of 2.7. This gap is further amplified in physical space, where the mean inter-player distance increases by a factor of 3.22, from 3.26 km to 10.50 km. The FP-greedy pursuer holds every evader to a lower game value than any non-learning pursuer does, while the FP-greedy evader extracts the highest game value against almost every opponent. The sole exception occurs against the FP-greedy pursuer, in which the FP-mixed evader sees benefits from randomization in increased distance.

Notably, the greedy, mixed, and random pursuer columns remain comparable to one another, suggesting a physical limit inherent to the pursuit task: because all six candidate vertices lie on a single reachability polygon, the penalty for selecting a non-Nash support trajectory is bounded by the

TABLE II: **Mean inter-player distance (km) at  $s = 30$ .** Strategy match-ups match the game value equivalent. Higher values distances favor the evader.

| P1 \ P2   | FP-greedy   | FP-mixed | greedy | mixed        | random |
|-----------|-------------|----------|--------|--------------|--------|
| FP-greedy | 4.39        | 6.04     | 10.24  | <b>10.50</b> | 8.22   |
| FP-mixed  | 4.52        | 5.63     | 7.43   | 7.74         | 7.25   |
| greedy    | <b>3.26</b> | 3.94     | 4.69   | 4.64         | 4.39   |
| mixed     | 3.61        | 4.14     | 4.72   | 4.77         | 4.66   |
| random    | 4.33        | 5.10     | 6.14   | 6.41         | 6.20   |

polygon diameter. Moreover, the greedy strategy collapses the pursuer’s mixed equilibrium into a deterministic pure strategy. The highest-weight trajectory in a minimax solution is not the one that most aggressively closes distance but the one that hedges against the evader’s best escape, minimizing the pursuer’s worst-case scenario rather than optimizing against any single evasion trajectory.

3) *Effect of Time on Learning*: The strategy ranking is stable across all three game step windows, but the game value heatmaps for 5, 10, and 30 game steps in Figure 3 reveals a time-dependent effect: the pursuer closes distance over successive orbits. The strategy pair rankings are more or less uniform at 5 game steps, although the FP-greedy evader quickly learns to exploit the predictable behavior of the greedy pursuer. The distribution sharpens at 10 game steps as the pursuer benefits from accumulated observations, universally deflating the game values. By 30 game steps, the rankings have largely stabilized, although the additional observations continue to benefit the pursuer. Nevertheless, across all game windows, the FP-strategies outperform the non-FP ones, demonstrating the benefits of learning over locally optimal but naive actions.

4) *Fuel Efficiency*: The  $\Delta V$  expenditure data reveal that the FP advantage is not obtained at the cost of additional fuel; in fact, the opposite is true. Figure 4 shows the relationship between  $\Delta V$  and mean distance (km), and Table III shows the cumulative  $\Delta V$  for three match-ups: FP-greedy vs. greedy, greedy vs. FP-greedy, and FP-greedy vs. FP-greedy.



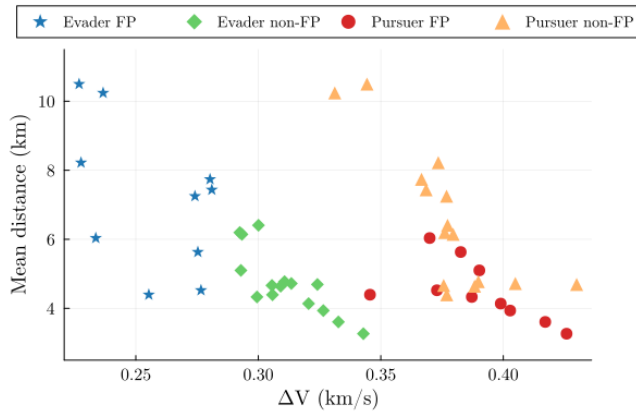


Fig. 4: **Evader and Pursuer  $\Delta V$  vs. Mean Distance (km).** The FP evader significantly decreases fuel expenditures while achieving higher mean distance, while the FP pursuer expends the same amount of fuel.

TABLE III: **Cumulative  $\Delta V$  (km/s) and mean distance for selected matchups at  $s = 30$ . E = Evader, P = Pursuer.**

| Matchup                         | E $\Delta V$ | P $\Delta V$ | P/E  | Distance |
|---------------------------------|--------------|--------------|------|----------|
| FP-greedy (E) vs. greedy (P)    | 0.237        | 0.331        | 1.40 | 10.24    |
| greedy (E) vs. FP-greedy (P)    | 0.343        | 0.426        | 1.24 | 3.26     |
| FP-greedy (E) vs. FP-greedy (P) | 0.255        | 0.346        | 1.36 | 4.39     |

The pursuer universally expends more fuel than the evader, as closing a gap against a non-cooperative target requires corrective maneuvers that compound over successive game steps. Additionally, the greedy evader spends 45% more  $\Delta V$  than the FP-greedy evader (0.343 vs. 0.237 km/s) while achieving only one-third the separation (3.26 vs. 10.24 km). The FP framework thus yields a dramatic improvement in fuel efficiency for the evader.

## VI. DISCUSSION

The results confirm the central thesis of this work: projecting the continuous orbital pursuit-evasion problem onto a discrete set of candidate trajectories and solving the resulting matrix game yields computationally tractable strategies that are also effective. The fictitious-play strategies consistently outperform deterministic and random baselines in both game value and fuel efficiency. The advantage is primarily informational: all players share the same action space and  $\Delta V$  constraints, yet FP players achieve better outcomes by learning the opponent’s behavior online. Moreover, even a uniform random evader outperforms a greedy evader at every horizon, confirming that unpredictability has inherent value: a deterministic strategy, no matter how locally optimal, is exploitable by a learning opponent.

In future work, the fictitious-play framework can be extended with Bayesian meta-strategy reasoning, in which the agent maintains a posterior belief over a library of opponent types and best-responds by marginalizing over this belief. This would enable the agent to distinguish not only *which*

vertices the opponent favors but *why*, and accelerate learning against dynamic strategies.

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